KEY QUESTIONS FOR AI-ENABLED MEDICAL DEVICE DEVELOPMENT

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EXECUTIVE SUMMARY

As the healthcare industry continues to integrate **Artificial Intelligence (AI) and Machine Learning (ML)** into medical devices, several aspects need to be carefully considered to ensure the reliability, safety, and security of these technologies. Below, we will cover critical questions that medical device manufacturers who are developing AI-Enabled medical devices should be asking themselves early on in their design process to ensure a more seamless experience.

Key topics include determining whether a device truly qualifies as an AI/ML medical device, understanding the specific problems AI is solving in clinical settings, and evolving risk management processes to address the unique challenges posed by AI-based software. It is essential to ensure that test data is representative of the intended patient population and maintain transparency and explainability in AI decision-making. Other important considerations include managing model drift, which can impact the accuracy of AI models over time and ensuring robust security throughout the AI development process. Finally, verifying and validating AI models is crucial to ensure their safety, efficacy, and reliability in real-world clinical applications.

By addressing these factors, we can better mitigate risks while maximizing the benefits of AI in medical devices and ensuring the highest quality of product is reaching patients and clinicians.

QUESTION 1: In the context of medical devices and software – What is AI?

Artificial Intelligence (AI) in medical devices and software refers to the use of computational techniques that enable machines to learn from data and make predictions, classifications, detect anomalies, or generate new data such as text or images.

AI in medical devices and software can take various forms, including:

- Machine Learning (ML), which learns patterns from data, often improving over time. ML models can be used for such things as anomaly detection and optimized decision-making supporting activities such as AI-assisted robotic surgery.
- Deep Learning (DL), which uses neural networks for complex tasks like medical imaging analysis enabling capabilities such as the detection of tumors in radiology scans with accuracy that is comparable to or exceeds human radiologists.
- Natural Language Processing (NLP), which processes text and speech for applications like clinical documentation.
- Generative AI (GenAI), such as LLMs create new data, such as synthetic medical images or AI-assisted reports.

AI in healthcare spans from simple rule-based systems to advanced deep learning models, each requiring different levels of validation and regulatory scrutiny. Identifying the optimal AI type early helps ensure the right design considerations, risk management, and compliance measures are incorporated in the product.

QUESTION 2: Does my device actually contain machine learning-enabled device software function(s)?

Establishing whether a device qualifies as a machine learning-enabled system is essential for regulatory strategy and compliance. It's a common misconception to assume that complex algorithms or statistical methods automatically equate to machine learning. However, true machine learning involves an adaptive model that "learns" and improves from data patterns over time, which subjects it to further scrutiny from regulators. Misclassifying a device as simple software when it's actually machine learning-enabled can lead to delays, additional risk classification, or rework during regulatory submission if the distinction isn't made early.

Determining if your device utilizes machine learning also has implications for regulatory requirements and risk classification. For example, in the U.S., the Food and Drug Administration (FDA) distinguishes between general **Device Software Functions (DSF)**, defined as any software function meeting the device definition under section 201(h) of the Food Drug & Cosmetic Act (FD&C Act), and **Machine Learning-Enabled Device Software Functions (ML-DSF)**, which integrate a trained ML model. The dynamic nature of ML-DSFs means they must undergo more rigorous validation, monitoring, and change control to ensure ongoing safety and efficacy. In the EU, software for medical purposes is categorized broadly as **Medical Device Software (MDSW)** under the Medical Device Regulation (MDR), meaning all software with medical intent must follow MDR requirements, though AI/ML systems may encounter further scrutiny under the EU AI Act.

International guidance highlights the importance of early identification of AI/ML components to align with specific regulatory standards. FDA's guidance on Predetermined Change Control Plans provides definitions for DSF and ML-DSF, clarifying how adaptive features introduce additional regulatory requirements. In the EU, Medical Device Coordination Group (MDCG) guidance outlines Medical Device Software (MDSW) regulations, while tools like the EU AI Act Compliance Checker help manufacturers navigate AI-specific considerations. Early classification ensures compliance with performance, transparency, and safety expectations across regulatory landscapes.

QUESTION 3: What problem is the AI solving in the clinical setting?

Identifying the clinical problem that an AI-enabled device addresses is the foundational step for any medical device development process. This question shapes the device's regulatory pathway and ensures that the AI application aligns with a specific clinical purpose, which is essential for defining intended use. For AI-enabled devices, a clearly defined clinical application not only sets a focused direction for development but also signals to regulators that the device meets a real healthcare need. For example, an AI may be developed reduce false positives in breast cancer screening.

Determining the intended clinical application early in development provides clarity on the compliance requirements for both regulatory bodies and quality management systems. In the U.S., understanding the intended clinical purpose helps establish whether the device aligns with an existing predicate or if it requires a novel classification through the de novo pathway. In the EU, it aids in determining the appropriate MDSW classification under MDR guidelines, which influences the rigor of technical evaluations, validation studies, and clinical performance assessments. Knowing the compliance requirements in advance makes the development process more predictable and aligns quality and risk management strategies.

Key guidance documents, such as the FDA's list of authorized AI/ML-enabled devices and the MDCG's guidance on Medical Device Software, provide essential resources for ensuring compliance. For U.S.-based manufacturers, the FDA's "Predetermined Change Control Plan for AI/ML-Enabled Software Functions" offers insight into maintaining regulatory compliance for devices with adaptive elements. In the EU, the MDR's guidance on software classification clarifies the regulatory pathway for AI applications in medical devices, supporting an efficient and compliant development process.

QUESTION 4: Are we tailoring our risk-management to account for the incorporation of AI in our software?

Risk management is a cornerstone of the medical device development process, but AI-enabled systems require a tailored approach that addresses **both safety risks and cybersecurity risks**. Traditional risk management frameworks focus heavily on safety—identifying hazards, mitigating risks, and ensuring patient safety. However, AI systems introduce additional complexities, including the dynamic nature of machine learning models, the reliance on data, and potential vulnerabilities to cyber threats. Failing to address these dimensions comprehensively can compromise both patient safety and system security.

From a **safety risk perspective**, AI systems must be evaluated for risks related to data lifecycle management, such as training data bias, overfitting, underfitting, and hallucinations. Performance drift—where AI accuracy degrades over time—is another significant concern, especially in adaptive systems. Bias in AI can lead to underdiagnosis in minority populations, while model drift can reduce accuracy over time, potentially leading to misdiagnoses. Tailoring risk management processes to include these AI-specific considerations ensures the device remains effective and safe across its lifecycle.

From a **cyber risk perspective**, AI-enabled systems are particularly vulnerable to threats such as adversarial attacks, data poisoning, and unauthorized access. Unlike static software, AI systems often rely on external data inputs, which increases exposure to malicious interference. Cybersecurity risk management must therefore address the integrity, availability, and confidentiality of AI-related data and algorithms, aligning with regulatory expectations and guidance on secure development.

To align with best practices, manufacturers can reference **AAMI TIR34971:2023**, which applies ISO 14971 principles specifically to machine learning systems. This standard ensures risks unique to AI are systematically identified, analyzed, and controlled. Additionally, resources like the **National Telecommunications and Information Administration's (NTIA) Requisites for AI Accountability and the**

Transparency for Machine Learning-Enabled Devices: Guiding Principles emphasize the need for robust accountability and transparency throughout the AI lifecycle. Tailoring risk management to include both safety and cybersecurity ensures AI systems are not only compliant but also trustworthy and reliable in clinical use.

QUESTION 5: Is our test data representative of the intended user population?

Ensuring that test data is representative of the intended user population is critical for developing robust, reliable, and effective AI systems. AI-enabled medical devices must perform consistently across the diverse populations they are intended to serve. If the user population is not appropriately defined or reflected in the test data, the device may exhibit bias, reduced performance, or even safety risks when deployed in real-world settings. Regulators increasingly focus on representativeness as a key element of AI transparency and fairness.

The first step is to **clearly define the intended user population**—including demographics, clinical conditions, geographic locations, and other relevant factors. This definition sets the foundation for robust testing and ensures that datasets include sufficient diversity to reflect the target users. Without this, AI systems risk underperforming or introducing harm to underrepresented populations. Furthermore, transparency about the user population and dataset composition is a cornerstone of regulatory compliance. As outlined in **Good Machine Learning Practices (GMLP)**, manufacturers must be able to justify their data sources and demonstrate how these datasets align with the intended user base.

Ensuring population representativeness not only meets regulatory expectations but also builds confidence in the device's performance and fairness. For AI systems, transparency about user population definition and test data helps mitigate bias, improves generalizability, and supports ongoing trust among regulators, clinicians, and patients. Following best practices like those outlined in GMLP ensures AI-enabled devices are evaluated rigorously and deliver consistent, equitable outcomes in real-world use.

QUESTION 6: How do we ensure transparency and explainability of AI decisions?

Transparency and explainability are critical for achieving regulatory approval and building trust with clinicians, patients, and customers. AI-enabled medical devices often operate as "black boxes," where decision-making processes are not immediately clear. Regulators and end-users alike require a clear understanding of how the system reaches its decisions, its performance characteristics, and its limitations to ensure the technology is safe, effective, and trustworthy.

Transparency begins with **clearly describing the decision-making process** and the data used to train, validate, and test the AI system. This includes disclosing relevant performance metrics, limitations of the model, and known risks such as biases or failure modes. For example, the FDA's **"Transparency for Machine**

Learning-Enabled Medical Devices: Guiding Principles" highlights the importance of communicating model inputs, outputs, and how variability in the data could impact results. This level of clarity ensures that clinicians and patients can make informed decisions based on AI recommendations.

Explainability goes one step further by providing insights into how the AI processes data to generate outputs. While not all systems can be fully interpretable, efforts to document and explain the AI's behavior foster trust in its reliability and performance. **Aligning with GMLP** helps manufacturers strike a balance between model complexity and transparency, ensuring that the AI is not only effective but also understandable to users. Demonstrating transparency is increasingly becoming a regulatory expectation and a competitive differentiator in the medical device market.

QUESTION 7: How are we accounting for model drift over time?

AI systems are susceptible to **model drift**, where their performance degrades over time as the input data or operating environment changes. Unlike traditional static systems, AI models depend on data patterns that can evolve, such as shifts in clinical practices, patient populations, or sensor inputs. Failing to account for drift can lead to reduced accuracy, reliability, and, ultimately, compromised patient safety or clinical efficacy.

To address drift effectively, it is essential to **establish baseline performance metrics** such as accuracy, precision, sensitivity, and specificity during the validation phase. These metrics serve as a benchmark for ongoing monitoring and testing. Continuous validation testing against real-world data and periodic evaluation using updated datasets help identify deviations from expected performance. For example, systematic drift can arise when input data begins to diverge significantly from the training or validation datasets, necessitating retraining or updates to the model.

Regulators and industry best practices emphasize the importance of **ongoing monitoring and robust change control processes** for AI systems. Establishing a baseline test during the development phase and continuously testing against newly collected data ensures that the system maintains its intended performance throughout its lifecycle. Resources like Microsoft's **AI Drift Overview** provide additional insights into identifying and managing drift, while regulatory frameworks, such as the FDA's guidance on AI/ML-enabled devices, stress the need for predetermined change control plans to address performance updates responsibly. Proactively managing model drift is not just about compliance—it's about ensuring that AI-enabled devices deliver consistent, reliable outcomes over time.

QUESTION 8: How are we addressing security throughout our product lifecycle?

Cybersecurity is a critical consideration for AI-enabled medical devices, as they are particularly vulnerable to cyberattacks that can compromise patient safety, data integrity, and device functionality. Unlike safety risks, which focus on unintended misuse, cybersecurity risks account for **foreseeable malicious intent**, such as adversarial attacks, data manipulation, and unauthorized system access. Regulators are increasingly scrutinizing "cyber devices" to ensure security measures are embedded throughout the development process rather than being treated as an afterthought.

While safety and cybersecurity risks are interconnected, they require **separate, tailored processes** to identify and mitigate hazards effectively. In safety risk management, "foreseeable misuse" is considered, but in cybersecurity, threats are often intentional and designed to exploit vulnerabilities. This evolving landscape demands a proactive approach, incorporating security-by-design principles, secure coding practices, and lifecycle risk management. Developers must implement robust safeguards such as encryption, access control, vulnerability testing, and incident response plans to mitigate both intentional and unintentional threats. AI model red teaming—systematically stress-testing AI models against adversarial threats—should also be part of this approach. This includes evaluating model resilience against data poisoning, adversarial inputs, and unauthorized modifications in order to detect and mitigate vulnerabilities before deployment, preventing adversarial manipulation.

Regulatory bodies have provided guidance to address these growing concerns. The FDA's **"Cybersecurity in Medical Devices: Quality System Considerations and Content of Premarket Submissions"** outlines expectations for integrating cybersecurity into the design, development, and post-market phases. In the EU, the **Medical Device Coordination Group (MDCG) Guidance on Cybersecurity for Medical Devices** reinforces similar principles, requiring manufacturers to demonstrate resilience against cyber threats throughout the device lifecycle. Additionally, the **International Medical Device Regulators Forum (IMDRF) Principles and Practices for Medical Device Cybersecurity** provide a global framework for aligning cybersecurity efforts with safety and performance requirements. Addressing cybersecurity risks with foresight, not hindsight, ensures compliance, protects patients, and builds trust in AI-enabled medical devices.

QUESTION 9: How do we verify and validate our AI model?

Formal verification and validation (V&V) are essential components of medical device testing and are critical for achieving regulatory approval. Verification ensures that the AI model meets its design specifications, while validation confirms that it performs effectively under real-world conditions. These steps provide objective evidence that the AI operates as intended and meets the performance requirements necessary for both regulatory compliance and clinical effectiveness.

Characterizing key performance metrics—such as **accuracy, precision, sensitivity, specificity, recall, and false positives**—is fundamental to establishing the reliability and robustness of AI systems. These metrics demonstrate how well the model performs its intended function, where its limitations may lie, and how it handles clinical edge cases. Additionally, if the AI system is designed to run on specific hardware, manufacturers must verify that the hardware can support the system's operation reliably over time. This is particularly important for ensuring consistency under varying conditions, such as sustained use in clinical environments or during software updates.

It's important to note that **all existing software validation guidances still apply** to AI-enabled systems. FDA resources, including the **"Content of Premarket Submissions for Device Software Functions"** and **"General Principles of Software Validation,"** provide comprehensive frameworks for establishing V&V processes. These guidances outline best practices for testing, traceability, and risk management throughout development. Incorporating these formal methods ensures the AI model is reliable, effective, and compliant with regulatory requirements, building confidence among regulators, clinicians, and end users.

QUESTION 10: What post-market surveillance processes do we have in place?

Post-market surveillance (PMS) is a regulatory requirement for all medical devices, but it is especially critical for AI-enabled systems. Unlike traditional static devices, AI systems have the potential to change over time due to their reliance on data inputs and adaptive learning processes. This dynamic nature introduces the risk of **model drift**—where performance degrades or deviates from its validated baseline—making proactive monitoring essential to ensure continued safety, efficacy, and compliance.

Effective post-market surveillance aligns with the **Total Product Lifecycle (TPLC)** approach, which emphasizes the need for continuous oversight after deployment. Tracking model performance in real-world use provides key insights into changes in accuracy, sensitivity, or specificity. These activities help manufacturers identify issues such as drift, bias, or changes in clinical outcomes early, allowing for corrective actions like retraining or updates to maintain device performance**. Integrating PMS processes with PCCPs**, as outlined in FDA guidance, ensures that any changes to the model are controlled and documented in a systematic way.

Regulatory bodies globally require post-market surveillance for medical devices. In the U.S., the **FDA's post-market requirements for devices** specify reporting obligations, including adverse events and corrective actions. Similarly, under the EU MDR, manufacturers must implement PMS plans that include ongoing performance monitoring, including the use of automated AI monitoring tools, and vigilance reporting. Guidance such as MDCG's **Post-Market Requirements for MDR** provides a clear framework for ensuring compliance in the EU market. For AI systems, enhanced post-market activities not only fulfill regulatory requirements but also safeguard patient safety, maintain trust, and support the ongoing reliability of the device.

QUESTION 11: Do we have a predetermined change control plan that is integrated with our QMS?

A PCCP is a critical tool for managing the evolving nature of AI-enabled medical devices. The FDA's recently finalized **Guidance on Predetermined Change Control Plans (2024)** highlights the need for manufacturers to proactively identify and document planned modifications to AI/ML systems, ensuring that changes maintain safety and effectiveness while remaining compliant. This approach integrates seamlessly into a device's existing **quality management system** (QMS), reducing the quality burden by aligning AI system control with established processes for risk management, testing, and post-market surveillance.

The PCCP provides a structured framework for managing AI model updates, including retraining, performance testing, and validation. AI-enabled systems are inherently dynamic, and without a clear plan to document and control changes, manufacturers risk introducing unintended performance issues, regulatory challenges, or non-compliance. By integrating the PCCP into the QMS, manufacturers can systematically address safety risk management, cybersecurity, and software testing—ensuring all changes to the AI system are tracked, validated, and communicated to regulators as necessary.

Beyond the FDA's guidance, this approach is likely to influence broader global harmonization efforts, making it a future-ready solution for manufacturers targeting international markets. Regulatory convergence initiatives, often driven by organizations like IMDRF, mean that frameworks such as **ISO 42001: Artificial intelligence – Management system** may soon align with PCCP principles, reinforcing its importance across jurisdictions. Integrating a PCCP with the QMS not only meets current FDA expectations but also provides manufacturers with a scalable, globally applicable process for controlling and maintaining AI-enabled systems throughout their lifecycle.

Key Takeaways

Developing AI-enabled medical devices requires manufacturers to navigate a litany of complex regulatory and technical considerations. From defining AI in the medical device context to having comprehensive AI-focused risk management to tailoring post market surveillance to the unique use of their product; each of these questions plays a crucial role in achieving regulatory approval and delivering safe, effective, AI-driven solutions to clinicians and patients.

As with any new technology, the regulatory and technical landscapes rapidly evolve. New guidance, standards, and expectations continue to emerge, reshaping how AI systems must be designed, analyzed, validated, and maintained. Failing to anticipate these shifts can lead to costly delays, compliance gaps, and unforeseen risks.

Our team of experts specializes in guiding manufacturers through this evolving environment, ensuring that every critical aspect—regulatory strategy, risk management, cybersecurity, testing, and lifecycle controls—is addressed proactively. Engaging with our consultants helps streamline your path to market, reduce regulatory uncertainty, and position your AI-enabled device for long-term success.

Links to Resources

Question	Topic	Resource Links
2	Medical Device Software & AI	- Medical Device Software (MDCG) - FDA Guidance: Predetermined Change Control for AI/ML - EU AI Act Compliance Checker - AI/ML in Software as a Medical Device
3	Authorized Devices & Classification	- FDA List of Authorized Devices - EU Guidance on Software Classification
4	AI Accountability & Transparency	- NTIA: AI Accountability Areas of Agreement - Transparency for ML-Enabled Devices - AAMI TIR34971:2023 - Good Machine Learning Practices (GMLP)
6	Transparency for Medical Devices	- FDA Transparency Principles for ML-Enabled Medical Devices
7	AI Model Drift	- Microsoft AI Drift Overview
8	Medical Device Cybersecurity	- FDA Guidance on Cybersecurity - MDCG Cybersecurity Guidance - IMDRF Principles for Medical Device Cybersecurity - Cyber Device Definition 524B(c)
9	Premarket Submissions & Validation	- Premarket Submission Content for Device Software - General Principles of Software Validation - Google Dev Course: Accuracy, Precision, Recall
10	Post-Market Requirements	- FDA Predetermined Change Control Plans for ML Devices - FDA Post-Market Requirements - MDCG Post-Market Requirements
11	AI Management & Guidance	- ISO 42001: AI Management System - FDA Predetermined Change Control Guidance (2024)

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